MySchola

GCSE BIOLOGY

TOPIC 1: CELL BIOLOGY

Detailed AQA Specification Notes

Clear explanations • Required practicals • Maths skills • Exam tips

HOW TO USE THIS BOOKLET

Learn each section, then use the exam tips and check yourself questions to test whether you can explain the science in full sentences. The content is scoped to the uploaded AQA Topic 1 specification; the explanations expand those points to an exam ready GCSE level.

MySchola • Learn clearly. Practise purposefully.

Topic map

Everything in AQA Topic 1: Cell Biology at a glance

Section	Focus	What you must be able to do
1	Cell types, scale and units	Eukaryotes, prokaryotes, prefixes, standard form
2	Animal, plant and bacterial cells	Structures and their functions
3	Specialised cells and differentiation	How cell structure links to function
4	Microscopy	Light vs electron microscopes, magnification, Required Practical 1
5	Culturing microorganisms	Biology only, aseptic technique, Required Practical 2
6	Chromosomes and mitosis	DNA, chromosomes, cell cycle and growth
7	Stem cells	Embryonic, adult and plant stem cells, benefits and risks
8	Diffusion and surface area : volume	Rates of diffusion and why exchange surfaces are needed
9	Exchange surfaces	Lungs, small intestine, fish gills, roots and leaves
10	Osmosis	Water movement, calculations, Required Practical 3
11	Active transport	Movement against the concentration gradient
12	Maths and exam checklist	Core equations, common errors and revision checklist

SPECIFICATION FLAGS

BIOLOGY ONLY means content that is not part of Combined Science. REQUIRED PRACTICAL means you must know the method, variables, safety and how to interpret results. HT ONLY is shown only where the uploaded specification explicitly labels a Higher Tier requirement.

1 Cell types, scale and units

Specification focus: eukaryotes, prokaryotes and cell size

Cells are the basic unit of life. AQA expects you to recognise two broad cell organisations: eukaryotic cells and prokaryotic cells.

Feature	Eukaryotic cells	Prokaryotic cells
Examples	Animal and plant cells	Bacterial cells
Nucleus	Genetic material is enclosed inside a nucleus	No nucleus. Genetic material is a single DNA loop in the cytoplasm
Cell membrane	Present	Present
Cytoplasm	Present	Present
Cell wall	Present in plant and algal cells; made of cellulose	Present around the cell membrane
Extra DNA	Chromosomes are inside the nucleus	May contain small rings of DNA called plasmids
Relative size	Larger	Much smaller

EXAM LANGUAGE

Do not write that bacteria have “no DNA”. Bacteria do have DNA, but it is not enclosed inside a nucleus. Their main DNA forms a loop and they may also contain plasmids.

Scale and prefixes

Prefix	Symbol	Meaning	Useful conversion
centi	c	one hundredth	1 cm = 10^{-2} m
milli	m	one thousandth	1 mm = 10^{-3} m
micro	µ	one millionth	1 µm = 10^{-6} m
nano	n	one billionth	1 nm = 10^{-9} m

Order of magnitude means comparing sizes by powers of ten. For example, an object that is 10^{-6} m is one thousand times larger than an object that is 10^{-9} m, because there is a difference of three powers of ten.

CHECK YOURSELF

A bacterial cell is 2 µm long. Write this in metres using standard form.

Answer: 2×10^{-6} m

2 Animal, plant and bacterial cells

Know each structure and link it to its function

Animal cell structures

Structure	Function
Nucleus	Contains the genetic material (DNA). It controls the activities of the cell by controlling which genes are used.
Cytoplasm	Jelly like material where many chemical reactions happen. It contains enzymes that control these reactions.
Cell membrane	Controls the movement of substances into and out of the cell.
Mitochondria	Site of aerobic respiration. Energy released by respiration is transferred for cell processes.
Ribosomes	Site of protein synthesis. Proteins are made here.

Plant and algal cells

Plant cells contain the structures above and also have additional structures:

Structure	Function
Chloroplasts	Contain chlorophyll, which absorbs light for photosynthesis.
Permanent vacuole	Contains cell sap. It helps maintain internal pressure and supports the cell.
Cell wall	Made of cellulose. It strengthens and supports the cell while allowing substances to pass through.

Bacterial cell features

- Cell membrane and cytoplasm are present.
- A cell wall surrounds the cell membrane.
- The main genetic material is a single circular DNA loop and is not enclosed in a nucleus.
- One or more plasmids may be present. Plasmids are small rings of DNA.

COMMON ERROR

A plant cell does not have “extra” structures instead of animal cell structures. It has the common eukaryotic structures plus a cellulose cell wall, chloroplasts in photosynthetic cells and a permanent vacuole.

CHECK YOURSELF

Why would a muscle cell usually contain many mitochondria?

Answer: Muscle contraction requires energy. More mitochondria allow a high rate of aerobic respiration and therefore more energy transfer.

3 Specialised cells and differentiation

Structure is adapted to function

A specialised cell has features that allow it to carry out a particular function efficiently. AQA can give you an unfamiliar cell and ask you to link its structures to its function, so always use the pattern: feature → effect → function.

Cell	Key adaptations	How the adaptations help
Sperm cell	Tail; many mitochondria; acrosome containing enzymes; streamlined shape	Tail propels the cell. Mitochondria release energy for movement. Acrosome enzymes help the sperm penetrate the egg.
Nerve cell	Very long axon; branched ends; insulating sheath	Long axon carries electrical impulses over long distances. Branches connect with other cells. Insulation reduces loss of the electrical signal.
Muscle cell	Many mitochondria; long fibres; stores of glycogen	Mitochondria release energy for contraction. Long fibres shorten to produce movement. Glycogen provides a store of glucose for respiration.
Root hair cell	Long hair like extension; thin wall; many mitochondria	Large surface area increases absorption of water and mineral ions. Mitochondria provide energy for active transport of mineral ions.
Xylem cell	Long hollow tubes; no end walls when mature; walls strengthened with lignin	Forms a continuous pathway for water and mineral ions. Lignin strengthens the tissue and helps prevent collapse.
Phloem cell	Long conducting tubes; sieve plates; companion cells have many mitochondria	Transports dissolved sugars through the plant. Companion cells provide energy for loading and unloading sugars.

Cell differentiation

- As an organism develops, cells differentiate to become different types of specialised cells.
- Most animal cells differentiate at an early stage of development.
- Many plant cells keep the ability to differentiate throughout life.
- In mature animals, cell division is mainly used for repair and replacement.
- During differentiation, a cell develops different sub cellular structures so it can perform its particular function.

EXAM TIP

If asked why differentiation is important, explain that multicellular organisms need different cell types to perform different jobs. Differentiation produces specialised cells with the structures needed for those jobs.

4 Microscopy

Magnification, resolution and Required Practical 1

Light microscopes use light to form images of cells. Electron microscopes have much higher magnification and resolving power, so they can reveal much smaller cell structures in finer detail.

Feature	Light microscope	Electron microscope
Magnification	Lower	Much higher
Resolution	Lower. Two close points may blur together	Much higher. Two close points can be seen as separate
Cell detail visible	Main cell structures and whole cells	Much finer sub cellular detail
Impact on biology	Allowed cells to be viewed and compared	Increased understanding of many sub cellular structures

KEY DIFFERENCE

Magnification tells you how many times bigger the image is than the real object. Resolution is the ability to distinguish two close points as separate. A microscope can magnify an image without necessarily increasing useful detail if its resolution is poor.

Magnification calculations

FORMULA

magnification = image size ÷ real size

Rearrangements: image size = magnification × real size | real size = image size ÷ magnification

- Always put image size and real size in the same units before calculating.
- A drawing or photograph can include a scale bar. Use the scale bar to calculate actual size or magnification.
- Give answers in standard form when appropriate.

Worked example

A cell image is 36 mm long. The real cell is 60 µm long. Convert 36 mm to 36,000 µm. Magnification = 36,000 ÷ 60 = ×600.

Required Practical 1: observing cells with a light microscope

1. Place a thin sample of cells on a clean microscope slide. For plant cells, a thin epidermal layer can be used.
2. Add a drop of water or an appropriate stain so structures are easier to see.
3. Lower a coverslip gently to reduce trapped air bubbles.
4. Start with the lowest power objective lens. Use the coarse focus, then fine focus.
5. Move to a higher power objective if more detail is needed and refocus carefully.
6. Draw the cells using clear single lines. Add labels and include a magnification or scale.

BIOLOGICAL DRAWING RULES

Use a sharp pencil, draw clear continuous lines, do not shade, draw structures in proportion, label with straight ruled lines and include a magnification or scale bar.

5 Culturing microorganisms

BIOLOGY ONLY • Aseptic technique • Required Practical 2

BIOLOGY ONLY
This section is specifically labelled “biology only” in the uploaded AQA specification.

Bacterial growth

- Bacteria reproduce by binary fission, a simple form of cell division.
- If nutrients and temperature are suitable, some bacteria can divide as often as once every 20 minutes.
- Bacteria can be grown in nutrient broth or as colonies on agar gel plates.

Population calculation

If every bacterium divides into two, the population doubles each generation. Number of generations = total time ÷ mean division time. Population = starting population × 2^n, where n is the number of generations.

Example: one bacterium divides every 20 minutes for 2 hours. 120 ÷ 20 = 6 generations. Population = 1 × 2^6 = 64 bacteria.

HT ONLY
For the bacterial population calculation, the uploaded specification explicitly states that Higher Tier students should be able to express the answer in standard form.

Aseptic technique: preventing contamination

Step	Why it matters
Sterilise Petri dishes and culture media before use	Kills unwanted microorganisms that could contaminate the culture.
Sterilise the inoculating loop by passing it through a flame	Kills microorganisms already on the loop before the sample is transferred.
Keep the Petri dish lid closed as much as possible	Reduces contamination from microorganisms in the air.
Secure the lid with adhesive tape and store the plate upside down	Reduces contamination and helps stop condensation dripping onto colonies.
Incubate school cultures at a maximum of 25°C	Reduces the risk of growing harmful pathogens that grow well around body temperature.

Required Practical 2: antiseptics or antibiotics

1. Prepare an uncontaminated bacterial lawn on sterile agar using aseptic technique.
2. Place sterile paper discs containing different antiseptics or antibiotics onto the agar. Include a control disc where appropriate.
3. Secure the lid, label the plate and incubate it upside down at no more than 25°C in a school laboratory.
4. Measure the diameter or radius of each clear zone of inhibition.
5. Compare treatments. A larger clear zone shows stronger inhibition of bacterial growth under those conditions.

MATHS SKILL
Area of a circular clear zone = πr^2. Measure the radius carefully and keep units consistent.

6 Chromosomes and mitosis

DNA, the cell cycle and identical daughter cells

Chromosomes

- The nucleus contains chromosomes made of DNA molecules.
- Each chromosome carries a large number of genes.
- In body cells, chromosomes are normally found in pairs.

The cell cycle

Cells divide in a sequence of stages called the cell cycle. You need to understand the three overall stages, but the AQA specification states that you do not need to know the individual phases within mitosis.

Stage	What happens
1. Growth and DNA replication	The cell grows and increases the number of sub cellular structures such as mitochondria and ribosomes. DNA replicates so there are two copies of each chromosome.
2. Mitosis	One set of chromosomes is pulled to each end of the cell. The nucleus divides.
3. Cell division	The cytoplasm and cell membrane divide, producing two genetically identical daughter cells.

WHY MITOSIS MATTERS

Mitosis produces genetically identical cells. In multicellular organisms it is important for growth, development, repair and replacement of damaged or worn out cells.

How to recognise mitosis in context

- An embryo growing from one cell into many cells.
- A cut healing because new skin cells are produced.
- A plant root or shoot growing because cells divide in meristems.
- Replacement of cells that have been damaged or worn out.

COMMON ERROR

Do not say that mitosis itself is DNA replication. DNA is copied before mitosis. During mitosis, the copied chromosomes are separated so each new nucleus receives an identical set.

CHECK YOURSELF

A cell begins with one copy of each chromosome. Why must the DNA replicate before mitosis?

Answer: So there are two copies of each chromosome available to separate into the two daughter cells, allowing both to receive an identical set of genetic information.

7 Stem cells

Undifferentiated cells with the ability to form other cell types

A stem cell is an undifferentiated cell that can produce more stem cells and can differentiate into other types of cell.

Source	What it can do	Uses / importance
Human embryo	Can be cloned and made to differentiate into most types of human cell	Potential to replace damaged cells and treat conditions such as diabetes or paralysis.
Adult bone marrow	Can form several types of cells, including blood cells	Can be used to replace damaged blood forming cells.
Plant meristems	Can differentiate into any type of plant cell throughout the plant's life	Allows rapid cloning of plants with desirable characteristics.

Therapeutic cloning

In therapeutic cloning, an embryo is produced with the same genes as the patient. Stem cells from the embryo are therefore genetically matched to the patient, reducing the chance that the patient's body will reject the cells.

Benefits and risks of stem cell use

Potential benefits	Potential risks / concerns
May replace damaged or non functioning cells	Stem cells may transfer viral infections.
May help treat serious conditions such as diabetes and paralysis	Some people object to the use of human embryos for ethical or religious reasons.
Patient matched cells can reduce rejection in therapeutic cloning	Treatments must be carefully controlled so cells develop in the intended way.

Plant cloning using meristems

- Rare species can be cloned to help protect them from extinction.
- Crop plants with useful features, such as disease resistance, can be cloned quickly and economically.
- Clones are genetically identical, so useful characteristics are preserved.

EVALUATION TIP

For a 4 or 6 mark stem cell question, give both sides. Explain a medical benefit, explain a practical risk, then include the ethical or religious concern before reaching a reasoned conclusion if the question asks you to evaluate.

8 Diffusion and surface area to volume ratio

Movement down a concentration gradient

Diffusion is the spreading out of particles in a gas or in solution, producing a net movement from an area of higher concentration to an area of lower concentration.

Examples in organisms

- Oxygen diffuses into the blood during gas exchange and then into respiring cells.
- Carbon dioxide diffuses out of respiring cells and from the blood into the lungs.
- Urea diffuses from cells into the blood plasma so it can be transported to the kidneys for excretion.

Factors that increase the rate of diffusion

Factor	Why it increases diffusion rate
Larger concentration gradient	There is a bigger difference in concentration, so net movement from high to low concentration is faster.
Higher temperature	Particles have more kinetic energy and move faster, so they spread out more quickly.
Larger membrane surface area	More particles can cross the membrane at the same time.

Surface area to volume ratio

Single celled organisms have a relatively large surface area compared with their volume. This means diffusion across the cell surface can supply enough substances and remove wastes quickly enough for the whole cell.

As organisms become larger, volume increases faster than surface area. Their surface area to volume ratio becomes smaller, so simple diffusion across the outer surface is not enough to meet the needs of all cells.

CALCULATION

Surface area : volume ratio = surface area ÷ volume. Compare ratios using the same units. A larger value means more surface area is available per unit of volume.

CHECK YOURSELF

Why do large multicellular organisms need specialised exchange surfaces and transport systems?

Answer: Their surface area to volume ratio is small and many cells are far from the body surface. Diffusion alone would be too slow to move enough substances, so specialised surfaces and transport systems are needed.

9 Exchange surfaces

Adaptations that make diffusion efficient

Efficient exchange surfaces share several general features:

- A large surface area, so more particles can cross at once.
- A thin membrane, giving a short diffusion distance.
- In animals, a good blood supply, which carries substances away and maintains steep concentration gradients.
- For gaseous exchange in animals, ventilation refreshes the air or water and maintains concentration gradients.

Examples you must be able to explain

Exchange surface	Main adaptations
Small intestine	Villi and microvilli provide a very large surface area. The lining is thin. A rich blood supply carries absorbed substances away, maintaining gradients.
Lungs	Millions of alveoli create a large surface area. Alveolar and capillary walls are thin. Dense capillary networks and ventilation maintain steep oxygen and carbon dioxide gradients.
Fish gills	Many filaments and lamellae create a large surface area. Thin surfaces and a good blood supply help diffusion. Water flowing across the gills maintains concentration gradients.
Plant roots	Root hair cells provide a large surface area for absorption of water and mineral ions from the soil.
Leaves	Broad, thin leaves provide a large surface area and short diffusion path. Stomata allow gas movement, while internal air spaces increase the area available for gas exchange.

HOW TO ANSWER “EXPLAIN THE ADAPTATION”

Avoid listing features without explaining them. Write the feature and then its consequence: “The alveoli have a large surface area, so more oxygen molecules can diffuse into the blood per second.”

CHECK YOURSELF

Explain why a thin exchange surface increases the rate of diffusion.

Answer: A thin surface gives particles a shorter diffusion distance, so they cross the membrane more quickly.

10 Osmosis

Diffusion of water through a partially permeable membrane

Osmosis is the diffusion of water from a dilute solution to a concentrated solution through a partially permeable membrane.

KEY IDEA

A dilute solution has a higher concentration of water. A concentrated solution has a lower concentration of water. Water therefore moves from the dilute side to the concentrated side across a partially permeable membrane.

Predicting changes in plant tissue

Outside solution	Net water movement	Effect on plant tissue
More dilute than the cell contents	Water enters the cells by osmosis	Mass increases and cells become more turgid.
Same effective concentration	No overall net movement of water	Little or no change in mass.
More concentrated than the cell contents	Water leaves the cells by osmosis	Mass decreases and cells become less turgid.

Required Practical 3: effect of concentration on plant tissue

1. Prepare equal sized pieces of plant tissue, for example potato cylinders. Blot them dry and measure their initial mass.
2. Place each piece into a known concentration of sugar or salt solution. Use a suitable range of concentrations.
3. Keep important control variables the same, such as tissue size, solution volume, temperature and time left in solution.
4. Remove the tissue after the same time, blot it dry in the same way and measure final mass.
5. Calculate the change and percentage change in mass for each concentration.
6. Plot concentration on the x axis and percentage change in mass on the y axis. The concentration where percentage change is zero is the point with no overall net movement of water.

PERCENTAGE CHANGE

percentage change in mass = ((final mass − initial mass) ÷ initial mass) × 100

A positive answer means mass gained. A negative answer means mass lost.

Worked example

Initial mass = 4.00 g, final mass = 4.60 g. Change = +0.60 g. Percentage change = (0.60 ÷ 4.00) × 100 = +15%.

PRACTICAL ERROR TO AVOID

Blot the tissue before weighing. Water left on the outside would artificially increase the measured mass and reduce the validity of the result.

11 Active transport

Moving substances against the concentration gradient

Active transport moves substances from a more dilute solution to a more concentrated solution, against the concentration gradient. It requires energy transferred by respiration.

Biological examples

Example	Why active transport is needed
Root hair cells	Mineral ions may be at a lower concentration in the soil than inside the root hair cell. The ions are moved into the cell against the concentration gradient. Energy from respiration is required.
Small intestine	After digestion, sugar molecules may need to be absorbed from a lower concentration in the gut into a higher concentration in the blood. Active transport allows this absorption to continue against the gradient.

Compare the three transport processes

Process	What moves	Direction	Membrane needed?	Energy from respiration?
Diffusion	Particles in gases or solutions	High concentration → low concentration	Not always	No
Osmosis	Water only	Dilute solution → concentrated solution	Yes, partially permeable	No
Active transport	Dissolved substances such as mineral ions or sugars	Low concentration → high concentration	Across cell membranes	Yes

EXAM LANGUAGE

The phrase “against the concentration gradient” is essential for active transport. For osmosis, mention water and a partially permeable membrane. For diffusion, describe net movement from higher to lower concentration.

CHECK YOURSELF

Why do root hair cells contain many mitochondria?

Answer: Mitochondria carry out aerobic respiration, transferring energy that is needed for active transport of mineral ions into the root hair cell.

12 Maths skills for Topic 1

Equations, units and method

Skill	Equation / method	Exam reminder
Magnification	magnification = image size ÷ real size	Convert both sizes into the same units first.
Real size	real size = image size ÷ magnification	Write the final unit clearly.
Image size	image size = magnification × real size	Check that the image should be larger than the real object.
Percentage change in mass	((final − initial) ÷ initial) × 100	Keep the sign. Positive = gain, negative = loss.
Bacterial generations	number of generations = total time ÷ division time	Use consistent time units.
Bacterial population	starting number × 2^n	HT: express very large answers in standard form when required.
Area of inhibition zone	area = πr^2	Radius is half the diameter. Include squared units.
Surface area : volume	surface area ÷ volume	Compare values using the same units.

Worked maths practice

1. An image is 48 mm long and the real cell is 80 µm long. Convert 48 mm to 48,000 µm. Magnification = 48,000 ÷ 80 = ×600.
2. A potato cylinder changes from 3.20 g to 2.72 g. Change = −0.48 g. Percentage change = (−0.48 ÷ 3.20) × 100 = −15%.
3. A bacterial population starts at 5 cells and doubles every 30 minutes for 3 hours. There are 180 ÷ 30 = 6 generations. Population = 5 × 2^6 = 320 cells.
4. A clear zone has a radius of 7 mm. Area = π × 7^2 = approximately 154 mm^2.

CALCULATOR CHECK

After every calculation, ask: Are the units correct? Is the answer sensible? Should I use standard form? Have I included a sign for percentage change?

13 Topic 1 exam checklist

Tick each statement only when you can explain it without notes

☐	Compare eukaryotic and prokaryotic cells, including the location of genetic material.
☐	Use centi, milli, micro and nano correctly and convert between units.
☐	Explain the functions of the main structures in animal, plant and bacterial cells.
☐	Explain how specialised cells are adapted for their functions.
☐	Explain the importance of cell differentiation in animals and plants.
☐	Compare light and electron microscopes using magnification and resolution.
☐	Calculate magnification, image size and real size.
☐	Describe Required Practical 1 and produce a suitable biological drawing.
☐	Describe aseptic technique and explain why each step is needed.
☐	Describe Required Practical 2 and calculate the area of a zone of inhibition.
☐	Calculate bacterial population growth from a mean division time.
☐	Describe chromosomes and the three overall stages of the cell cycle.
☐	Explain why mitosis is important for growth, development, repair and replacement.
☐	Describe embryonic, adult and plant stem cells and evaluate their uses.
☐	Define diffusion and explain the factors that affect its rate.
☐	Calculate and compare surface area to volume ratios.
☐	Explain why large multicellular organisms need exchange surfaces and transport systems.
☐	Explain adaptations of the lungs, small intestine, fish gills, roots and leaves.
☐	Define osmosis and predict movement of water in different concentrations.
☐	Describe Required Practical 3 and calculate percentage change in mass.
☐	Define active transport and explain examples in roots and the small intestine.
☐	Compare diffusion, osmosis and active transport accurately.

FINAL EXAM RULE

Definitions alone are rarely enough for higher mark questions. Link cause to effect using precise biological vocabulary: concentration gradient, surface area, diffusion distance, respiration, energy, differentiation and genetically identical.

14 Exam style practice

Use these after revising the full topic

Q	Question	Marks
1	State two differences between a bacterial cell and an animal cell.	2 marks
2	Explain how the structure of a root hair cell is adapted for absorption of water and mineral ions.	4 marks
3	An image of a cell is 72 mm long. The actual cell is 120 µm long. Calculate the magnification.	3 marks
4	Describe the stages of the cell cycle that produce two genetically identical cells.	4 marks
5	Evaluate the use of embryonic stem cells in medical treatment.	6 marks
6	Explain why a multicellular organism needs specialised exchange surfaces.	4 marks
7	A potato cylinder has an initial mass of 5.0 g and a final mass of 4.2 g. Calculate the percentage change in mass.	3 marks
8	Compare diffusion, osmosis and active transport.	6 marks
9	Describe how to prepare and incubate an uncontaminated culture of bacteria in a school laboratory.	6 marks
10	Explain how the lungs are adapted for rapid gas exchange.	4 marks

MARK SCHEME MINDSET

For “explain” questions, link every scientific feature to a consequence. For practical questions, include the method, key control variables, safety or contamination controls and how results are measured. For “evaluate”, include benefits, risks and a justified judgement.

End of Topic 1: Cell Biology

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